Assignment 5: SQL Advanced Queries

Overview:

Build on your SQL skills by creating a products table and writing targeted queries. You'll practice filtering, pattern matching, and range conditions to extract insights from product data.

1. Create and Populate Table

```sql
CREATE TABLE Products (
  ProductID INT PRIMARY KEY,
  ProductName VARCHAR(50),
  Category VARCHAR(50),
  Price DECIMAL(10,2),
  Stock INT
);

INSERT INTO Products (ProductID, ProductName, Category, Price, Stock) VALUES
  (1, 'Laptop',  'Electronics', 800.00,  50),
  (2, 'Smartphone','Electronics', 600.00,  30),
  (3, 'Desk Chair','Furniture',   120.00, 100),
  (4, 'Table',    'Furniture',   200.00,  20),
  (5, 'Notebook',  'Stationery', 5.00, 500),
  (6, 'Pen',      'Stationery',     2.00,1000);
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_1)** and save it in the same **PDF/Doc/Word** file.

2. SQL Query Tasks

1. **a) Electronics & Furniture:** Retrieve products in categories Electronics or Furniture.
SQL Pattern: WHERE Category IN ('Electronics','Furniture')

Submission Type : *Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2a)** and save it in the same **PDF/Doc/Word** file.*

2. **b) Price Range:** Find products with price between 100 and 800.
SQL Pattern: WHERE Price BETWEEN 100 AND 800

Submission Type : *Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2b)** and save it in the same **PDF/Doc/Word** file.*

3. **c) Stock Range:** List products whose stock is between 50 and 500.
SQL Pattern: WHERE Stock BETWEEN 50 AND 500

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2c)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

4. **d) Name Contains 'Pen':** Select products with 'Pen' anywhere in the name.
SQL Pattern: WHERE ProductName LIKE '%Pen%'

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2d)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

5. **e) Name Starts With 'S':** Retrieve products whose names start with 'S'.
SQL Pattern: WHERE ProductName LIKE 'S%'

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2e)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

6. **f) Category & Price:** Products in Stationery or Furniture and price 100–300.
SQL Pattern: WHERE Category IN ('Stationery','Furniture') AND Price BETWEEN 100 AND 300

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2f)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

7. **g) Low-Price 'o':** Products priced 1–10 and name contains 'o'.
SQL Pattern: WHERE Price BETWEEN 1 AND 10 AND ProductName LIKE '%o%'

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2g)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

> Tip: If you encounter a "table already exists" error, rename your table (e.g., Products1) before running the CREATE statement.

Submission Guidelines:

Create One Main Folder (named Assignment5_YourName) which includes -

- One **Doc/PDF/Word** file containing all the Queries and the screenshots of each part.
- Put your SQL file as **Assignment5_YourName.sql**

Convert the folder into ZIP format.
Upload the ZIP folder on the submission portal.

Watch this to learn how to create ZIP file

https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact the Tutedude mentor.